

# Model Report: RAG Retrieval + Reranking (Model of Choice)

Project: Smart MCQ Solver Challenge

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## 1. Motivation

With a from-scratch model (Bi-LSTM, MAP@3 = 0.773) and a fine-tuned pretrained model (BERT+LoRA, MAP@3 = 0.683) already built, this third model takes a fundamentally different approach: **retrieval-augmented reranking**, using a pretrained sentence embedder in a zero-shot capacity (no fine-tuning) combined with a lightweight trained fusion layer.

The design directly follows up on two earlier findings:

- **TF-IDF's failure** showed that comparing a question's own text to its own options isn't very informative here, since distractors are near-paraphrases of the correct answer.
- **The EDA found that 293 test questions have a near-identical "twin" question in train**, just reworded with different instruction-wrapper phrasing.

This motivates retrieving similar *already-answered* questions from the training set and using their answers as external context — a genuine RAG pattern (retrieve → augment → predict), with the training set itself serving as the knowledge base since no external corpus was provided with the competition.

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## 2. Architecture

**No gradient-trained neural network.** The only trained component is a 2-feature logistic regression; everything else is pretrained/zero-shot or a fixed retrieval mechanism.

Knowledge base construction:

`train_split` questions + correct answers  $\rightarrow$  embed (all-MiniLM-L6-v2, zero-shot)  $\rightarrow$  FAISS index (IndexFlatIP, cosine similarity)

For each question to answer:

1. Embed the question, retrieve top-3 most similar past questions from the index

2. `retrieved_context` = weighted average of the top-3 retrieved answers' embeddings  
(weighted by retrieval similarity)

For each of the 5 candidate options:

3. `direct_sim` = cosine(embed(prompt), embed(option))

4. `retrieval_sim` = cosine(retrieved\_context, embed(option))

5. `fused_score` = LogisticRegression([`direct_sim`, `retrieval_sim`])

6. Rank the 5 options by `fused_score`  $\rightarrow$  top-3 for MAP@3

| Component                   | Detail                                                                                                                       |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Embedder                    | <code>all-MiniLM-L6-v2</code> ( <code>sentence-transformers</code> ), pretrained, <b>not fine-tuned</b>                      |
| Vector index                | FAISS ( <code>IndexFlatIP</code> ) over L2-normalized embeddings (inner product = cosine similarity)                         |
| Retrieval                   | Top-3 nearest past questions per query                                                                                       |
| Reranker                    | <code>LogisticRegression(class_weight="balanced")</code> on 2 features: <code>direct_sim</code> , <code>retrieval_sim</code> |
| Knowledge base (validation) | <code>train_split</code> only — never <code>valid_split</code> , to prevent leakage                                          |
| Knowledge base (test)       | Full <code>train.csv</code> (no leakage concern, since <code>test.csv</code> is a genuinely separate file)                   |

### 3. Leakage Safeguards

This model needed **two separate leakage protections**, one inherited and one new:

1. **Same** (`StratifiedGroupKFold`) **split** (grouped by core question, wrapper phrasing stripped) used for the Bi-LSTM and BERT+LoRA models, so no near-duplicate question crosses train/validation.

2. **RAG-specific guard:** the retrieval index used to evaluate `valid_split` is built exclusively from `train_split`. Without this, the model could trivially retrieve its own duplicate answer during validation and "solve" the question by lookup rather than genuine retrieval-augmented reasoning — verified programmatically (zero core-prompt overlap) before trusting any result, same protocol as the other two models.

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## 4. Results (Leakage-Free Validation Split, n=417)

### Headline metrics

| Metric                      | Score  |
|-----------------------------|--------|
| Validation Accuracy (Top-1) | 0.8153 |
| Validation MAP@3            | 0.8773 |

### Classification Report

| Class        | Precision | Recall | F1   | Support |
|--------------|-----------|--------|------|---------|
| A            | 0.75      | 1.00   | 0.86 | 69      |
| B            | 0.91      | 0.72   | 0.80 | 68      |
| C            | 0.83      | 0.80   | 0.81 | 132     |
| D            | 0.84      | 0.69   | 0.76 | 100     |
| E            | 0.77      | 1.00   | 0.87 | 48      |
| Accuracy     |           |        | 0.82 | 417     |
| Macro avg    | 0.82      | 0.84   | 0.82 | 417     |
| Weighted avg | 0.82      | 0.82   | 0.81 | 417     |

### Confusion Matrix

**A and E are solved almost perfectly (100% recall each)** — every true-A and true-E question is correctly identified. **D is the weakest class (69% recall)**, mostly confused with A (16 of 100 true-D questions predicted A) and C (13 confused with C). This is a marked improvement over BERT+LoRA, which struggled specifically with class A (30% recall there vs. 100% here) — the retrieval signal appears to resolve exactly the confusion pattern that

direct-similarity-only approaches (TF-IDF, and to a lesser extent BERT+LoRA) struggled with.

## 5. Reranker Analysis — the Key Finding

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Reranker coefficients: direct_sim = -2.43, retrieval_sim = 26.80
Reranker intercept: -24.08
```

The logistic regression learned to **almost entirely discount** `direct_sim` (even slightly negative weight) and **rely overwhelmingly on** `retrieval_sim` (11x larger magnitude). This is a strong, data-driven confirmation of the hypothesis first raised in the TF-IDF report: **comparing an option directly to its own question's text is a weak signal in this dataset**, because distractors are deliberately close paraphrases of the correct answer. What actually works is comparing each option to the *answers of similar past questions* — the retrieved context carries the discriminative signal that the question's own text does not.

### Signal distributions

Both `direct_sim` and `retrieval_sim` show substantial visual overlap between correct (green) and distractor (red) options when plotted as pooled histograms across the whole validation set. This might look at first glance like *neither* signal should work well — but this is a case where **the aggregate distribution is not the full story**. The reranker's job is to identify, within each individual question, the *relative* ranking among 5 options, not to apply a fixed global threshold. A signal can look weak in a pooled histogram (because "distractor" always outnumbers "correct" 4:1, and both classes span a similar range) while still being highly informative for the *within-question* comparison that actually drives ranking. This is exactly what the strong MAP@3 (0.877) despite the histogram overlap demonstrates, and is a useful nuance to raise explicitly in the report/viva: aggregate diagnostic plots and actual task performance can diverge when the task is fundamentally about *relative* ranking rather than *absolute* classification.

## 6. Comparison Across All Three Required Models

| Model                         | Type                      | Validation<br>MAP@3 | Validation Accuracy<br>(Top-1) |
|-------------------------------|---------------------------|---------------------|--------------------------------|
| Random baseline               | —                         | ≈0.367              | 0.20                           |
| TF-IDF + Cosine<br>Similarity | (diagnostic baseline)     | 0.280               | 0.158                          |
| Bi-LSTM Siamese<br>Encoder    | <b>Built from scratch</b> | 0.773               | 0.713                          |

| Model                       | Type                    | Validation<br>MAP@3 | Validation Accuracy<br>(Top-1) |
|-----------------------------|-------------------------|---------------------|--------------------------------|
| BERT-base-uncased +<br>LoRA | Pretrained (fine-tuned) | 0.683               | 0.542                          |
| RAG Retrieval + Rerank      | Model of choice         | 0.877               | 0.815                          |

The RAG model is the best performer in the entire project, despite having the least amount of trained parameters (2 logistic-regression weights + 1 intercept) and doing no gradient-based fine-tuning at all. This is a genuinely useful result for the report's overall conclusions section: for this specific dataset — where the central difficulty is that distractors are near-paraphrases of correct answers — **retrieval-augmented context from similar solved examples is more valuable than either learning better internal representations from scratch (Bi-LSTM) or adapting a large pretrained model's own representations (BERT+LoRA).**

### 7. Why This Model Satisfies the "Model of Choice" / RAG Requirement

- Implements the full retrieve → augment → predict RAG pattern: loading a vector database, retrieving external context based on the question prompt, and feeding the retrieved context + prompt + choices into the model.
- Uses a pretrained embedding model in a genuinely zero-shot capacity — distinct in character from both the from-scratch Bi-LSTM (fully trained representations) and the fine-tuned BERT+LoRA (adapted pretrained representations).
- Has valid W&B tracking, a leakage-safe evaluation protocol consistent with the other two required models, and a properly trained (if small) predictive component, satisfying the "at least three runs compared using common metrics" L1 viva checkpoint.

### 8. Limitations & Next Steps

- **Only 2 features** feed the reranker — adding more (e.g. option length, rank of `retrieval_sim` among the question's own 5 options, number of retrieved neighbors above a similarity threshold) could push performance further.
- **Retrieval quality depends entirely on train-set coverage** — questions whose "twin" doesn't exist in train (i.e., not part of the 293 overlapping core prompts) rely on the retrieved neighbors being merely topically related rather than near-duplicates; worth checking whether performance is bimodal (very high MAP@3 for questions with an exact train twin, versus lower for questions without one) as a natural next error-analysis step.

- **Try a larger/stronger embedder** (e.g. `all-mpnet-base-v2`) to see if retrieval quality improves further.
- **Ensemble candidate:** given this model's very different error pattern from the Bi-LSTM and BERT+LoRA (e.g. perfect A/E recall vs. BERT+LoRA's weak A recall), combining predictions across all three models is a strong candidate for further score improvement (Milestone 5: Ensembling).

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## 9. Artifacts

| File                                                  | Description                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------|
| <code>06_rag_retrieval_rerank.py</code>               | Full retrieval + reranking pipeline script                                     |
| <code>rag_reranker.joblib</code>                      | Saved logistic regression reranker + config                                    |
| <code>submission_rag.csv</code>                       | Kaggle-ready submission, format-validated                                      |
| <code>eda_outputs/rag_confusion_matrix.png</code>     | Validation confusion matrix (top-1)                                            |
| <code>eda_outputs/rag_signal_distributions.png</code> | Direct vs. retrieval similarity distributions                                  |
| W&B run                                               | Project <code>23f3003691-t22026</code> , run <code>RAG_Retrieval_Rerank</code> |

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## 10. Summary for the Overall Project Report

Across all three required models, the project's central narrative holds up end-to-end:

1. **TF-IDF (lexical similarity)** fails because this dataset's distractors are designed to be lexically close to the correct answer.
2. **Bi-LSTM (learned, from-scratch representations)** succeeds by learning task-specific patterns beyond lexical overlap.
3. **BERT+LoRA (adapted pretrained representations)** underperforms the from-scratch model, likely due to LoRA's limited adaptation capacity on a small dataset.
4. **RAG retrieval + reranking (zero-shot embeddings + retrieved context)** wins overall by sidestepping the core problem entirely — instead of trying to judge an option's correctness from the question's own text, it judges options against **evidence from similar already-solved questions**, which turns out to be the most reliable signal available in this dataset.